CE 766: Watershed Management (Fall 2026)

Data and Preliminary Results

Reformulating the Lake Problem for Powai Lake, Mumbai

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Summary of progress

- Collected 25 years of IMD gridded rainfall (2001–2025), lake morphometry, measured phosphorus and sewage data for Powai Lake.
- Built a water and phosphorus mass balance. It reproduces the observed lake phosphorus (predicted 1.21 mg/L vs observed 0.93 mg/L).
- Tested four sewage-diversion scenarios: lake phosphorus falls by 55–85% but stays above the hypereutrophic limit (0.1 mg/L) in most cases.
- Estimated the Lake-Problem parameters. As a result, the model time step will change from annual to monthly.

1. Data collected

Table 1: Datasets used so far.

Data	Source	Period	Used for
Daily gridded rainfall, 0.25°	IMD [2]	2001–2025	Water balance, rainfall variability
Lake area $A_L = 2.2\,\text{km}^2$, catchment $A_C = 6.61\,\text{km}^2$, depth 3–10 m	Lekshmi et al. [3]	—	Lake volume, residence time
Total phosphorus (TP)	Lekshmi et al. [3]	Apr 2018 – Mar 2020	Checking the model
Phosphate	Usman et al. [4]	Nov 2016 – Oct 2017	Seasonal pattern
Sewage inflow (≈ 18 MLD) and diversion plan	BMC, via news reports [9]	2025–2026	Loads and scenarios

Rainfall at Powai. The IMD yearly binary files were read directly. Powai (19.13° N, 72.91° E) lies between four grid points, so daily rainfall was obtained by bilinear interpolation:

$$P_{\text{Powai}} = \sum_{i=1}^{4} w_i\, P_i, \qquad \varphi = \frac{19.13 - 19.00}{0.25} = 0.52, \qquad \lambda = \frac{72.91 - 72.75}{0.25} = 0.64 \tag{1.1}$$

where the weights are $w_1 = (1-\varphi)(1-\lambda)$, $w_2 = (1-\varphi)\lambda$, $w_3 = \varphi(1-\lambda)$ and $w_4 = \varphi\lambda$.

2. Method and formulas

2.1 Annual water balance

The outflow from the lake is the catchment runoff plus rain on the lake, minus evaporation, plus wastewater inflow:

$$Q = P\,C\,(A_C - A_L) \;+\; P\,A_L \;-\; E\,A_L \;+\; Q_{\text{ww}} \tag{2.1}$$

where P is annual rainfall (m), C the runoff coefficient, E lake evaporation (m/yr) and Q_{ww} the wastewater inflow (m³/yr).

2.2 Phosphorus load

$$L = Q_{\text{ww}}\,\text{TP}_{\text{ww}} \;+\; e_u\,(A_C - A_L) \;+\; d\,A_L \tag{2.2}$$

where TP_{ww} is the phosphorus concentration of the wastewater, e_u the urban export coefficient and d the atmospheric deposition on the lake (both in kg/ha/yr).

2.3 Residence time, retention and lake phosphorus

$$V = A_L\,\bar{z}, \qquad \tau = \frac{V}{Q} \tag{2.3}$$

$$R = \frac{1}{1 + \sqrt{\tau}} \qquad \text{(Larsen–Mercier retention [5])} \tag{2.4}$$

$$\text{TP}_{\text{lake}} = \frac{L}{Q}\,(1 - R) \tag{2.5}$$

Here $\bar{z}$ is the mean depth, τ the residence time (yr) and R the fraction of incoming phosphorus retained in the sediments.

2.4 Link to the Lake Problem

The phosphorus loss rate (flushing plus settling) and the equivalent Lake-Problem parameter b are:

$$k = \frac{Q/V}{1 - R} = \frac{1}{\tau\,(1 - R)}, \qquad b_{\text{annual}} = 1 - e^{-k}, \qquad b_{\text{monthly}} = 1 - e^{-k/12} \tag{2.6}$$

The classroom lake model (Carpenter et al. [1]) is

$$X_{t+1} = X_t + a_t - b\,X_t + \frac{X_t^{\,q}}{1 + X_t^{\,q}} + \varepsilon_t \tag{2.7}$$

At equilibrium ($X_{t+1} = X_t$, no noise) the load needed to hold the lake at level X is

$$a(X) = b\,X - \frac{X^q}{1 + X^q} \tag{2.8}$$

The lake type then follows from two thresholds:

$$g_{\max} = \max_X \frac{X^{q-1}}{1+X^q} = \frac{(q-1)^{(q-1)/q}}{q} \tag{2.9}$$

$$f'_{\max} = \max_X \frac{q\,X^{q-1}}{\left(1+X^q\right)^2} \tag{2.10}$$

Table 2: Lake classification rules.

Condition	Lake type
$b < g_{\max}$	Irreversible (cannot recover even with zero load)
$g_{\max} \le b < f'_{\max}$	Hysteretic (recovers only after a deep load cut)
$b \ge f'_{\max}$	Reversible (responds smoothly to load changes)

For a real lake, b is the loss rate relative to the maximum sediment recycling rate r and the half-saturation level m: $b = k\,m/r$. Powai therefore becomes hysteretic only if $r/m > k/f'_{\max}$, and irreversible only if $r/m > k/g_{\max}$.

2.5 Monte Carlo setup

Inputs that have not been measured were sampled from the ranges in Table 3 ($N = 20{,}000$ samples).

Table 3: Uncertain inputs used in the Monte Carlo analysis.

Symbol	Input	Range	Source
P	Annual rainfall	1,938 – 4,273 mm (resampled)	IMD [2]
$\bar{z}$	Mean lake depth	2.5 – 5.0 m	Assumed from [3]
C	Runoff coefficient	0.4 – 0.8	Assumed
E	Lake evaporation	1.2 – 1.8 m/yr	Assumed
Q_{raw}	Raw sewage inflow (before diversion)	15 – 21 MLD	[9]
TP_{raw}	TP of raw sewage	4 – 12 mg/L (mode 7)	[6]
e_u	Urban TP export coefficient	0.5 – 2.5 kg/ha/yr	[7]
d	Atmospheric deposition	0.1 – 0.5 kg/ha/yr	Assumed
f	Raw sewage leaking in after diversion	0 – 15%	Assumed
TP_{sec} / TP_{ter}	TP of secondary / tertiary STP effluent	3 – 6 / 0.5 – 1.0 mg/L	Assumed

3. Preliminary results

3.1 Rainfall

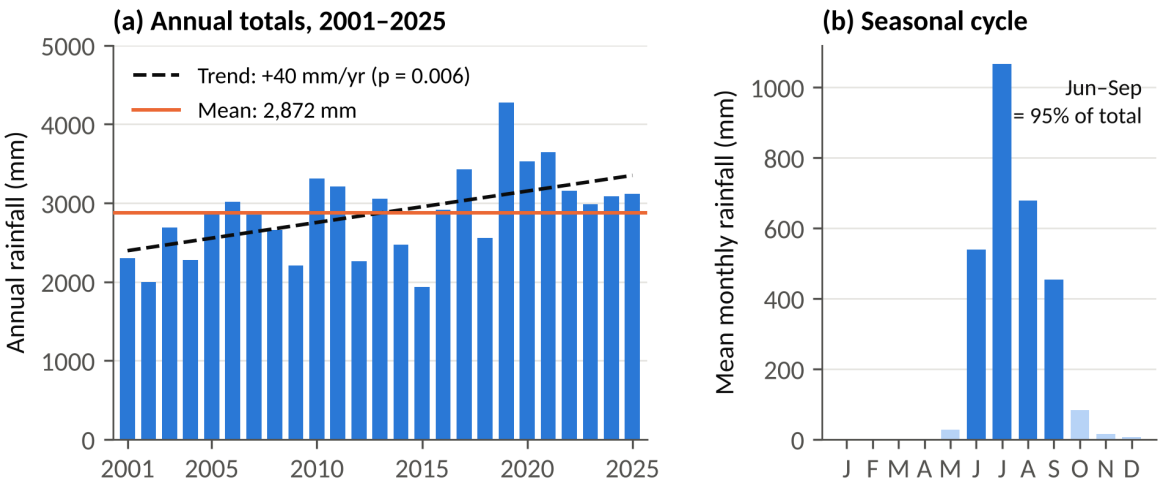


Figure 1: IMD rainfall at Powai: (a) annual totals with mean and linear trend; (b) average monthly rainfall.

Table 4: Rainfall statistics, 2001–2025.

Statistic	Value
Mean annual rainfall	2,872 mm
Standard deviation (coefficient of variation)	553 mm (0.19)
Driest / wettest year	1,938 mm (2015) / 4,273 mm (2019)
Share of rain in June–September	95%
Lognormal fit, ln $P \sim \mathcal{N}(\mu, \sigma^2)$	median 2,821 mm, $\sigma = 0.19$
Shapiro–Wilk test on ln P	$p = 0.81$ (lognormal not rejected)
Linear trend	$+40$ mm/yr ($p = 0.006$)

The lognormal fit supports the course's lognormal natural-inflow assumption. However, the significant upward trend means the rainfall distribution is not stationary, so it should be treated as a deep uncertainty.

3.2 Mass balance check (before diversion)

Worked example using median values

$$\frac{L}{Q} = \frac{50.1\,\text{t/yr}}{17.0\,\text{Mm}^3/\text{yr}} = 2.94\,\text{mg/L}$$

$$R = \frac{1}{1+\sqrt{0.48}} = 0.59$$

$$\text{TP}_{\text{lake}} = 2.94 \times (1-0.59) = \mathbf{1.20}\,\text{mg/L} \qquad \text{(observed: } 0.93 \pm 0.61\,\text{mg/L)}$$

$$k = \frac{1}{0.48 \times (1-0.59)} = 5.1\,\text{yr}^{-1}$$

Table 5: Mass balance results before diversion (median, with 5–95% range).

Quantity	Symbol	Value
Lake volume	V	8.2 Mm³ (5.8 – 10.7)
Outflow	Q	17.0 Mm³/yr
Residence time	τ	0.48 yr (0.31 – 0.74)
Phosphorus retention	R	0.59
External phosphorus load	L	50 t/yr (33 – 73)
Share of load from sewage	—	99%
Predicted lake TP	TP_{lake}	1.21 mg/L (0.70 – 2.03)
Observed lake TP, 2018–20 [3]	—	0.93 ± 0.61 mg/L

The observed value lies inside the predicted range, so the simple model reproduces the lake's present state without any calibration.

3.3 Sewage-diversion scenarios

The four scenarios differ only in what still enters the lake:

$$\begin{aligned}
\text{S0 (before):} \quad & L_{\text{ww}} = Q_{\text{raw}}\,\text{TP}_{\text{raw}} \\
\text{S1 / S2 (STP return):} \quad & L_{\text{ww}} = Q_{\text{STP}}\,\text{TP}_{\text{sec/ter}} + f\,Q_{\text{raw}}\,\text{TP}_{\text{raw}}, \qquad Q_{\text{STP}} = 8\,\text{MLD} \\
\text{S3 (no return):} \quad & L_{\text{ww}} = f\,Q_{\text{raw}}\,\text{TP}_{\text{raw}}
\end{aligned}$$

Table 6: Steady-state lake TP for each scenario (median, with 5–95% range).

Scenario	Load L (t/yr)	Lake TP (mg/L)	Chance TP $>$ 0.1 mg/L
S0: Before diversion	50.1 (33 – 73)	1.21 (0.70 – 2.03)	100%
S1: Diversion + secondary effluent	17.6 (12 – 24)	0.55 (0.31 – 0.96)	100%
S2: Diversion + tertiary effluent	6.4 (3.2 – 11)	0.20 (0.09 – 0.41)	92%
S3: Diversion, no effluent return	4.2 (1.1 – 8.9)	0.18 (0.04 – 0.46)	76%

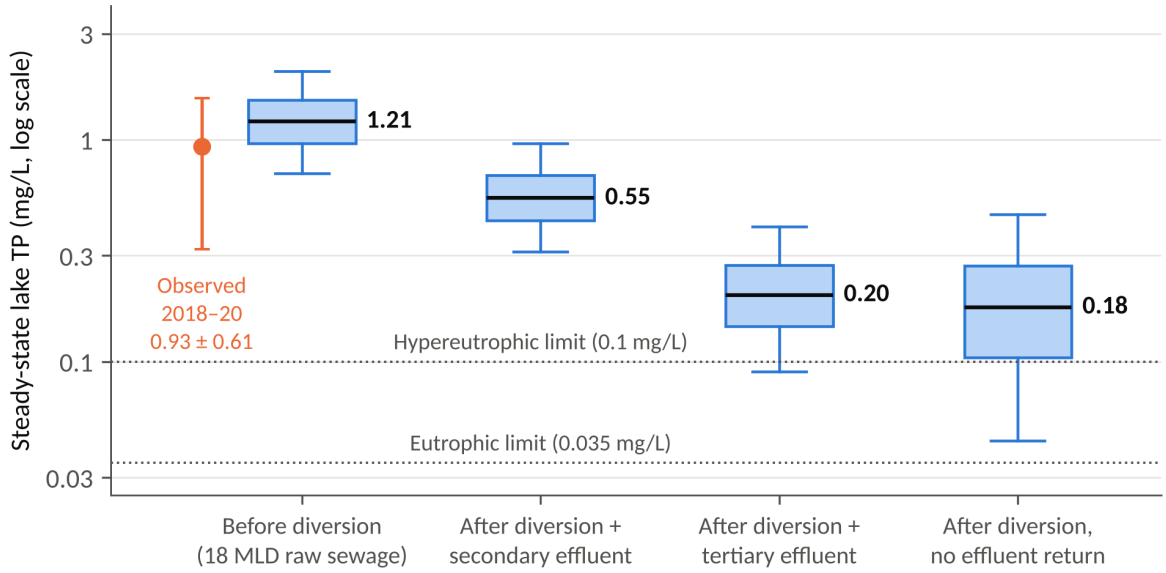


Figure 2: Predicted steady-state lake TP for each scenario (box: 25–75%; whiskers: 5–95%; number: median). Orange: observed 2018–20 TP (mean ± SD) [3]. Dotted lines: OECD trophic limits [8].

Diversion lowers lake phosphorus by 55–85%, but the lake stays hypereutrophic in most runs. Returning treated effluent keeps phosphorus high unless the STP removes phosphorus (tertiary treatment).

3.4 Which uncertainty matters most?

Table 7: Spearman rank correlation ρ_s between each input and lake TP (scenario S2).

Input	ρ_s
Raw sewage still leaking in, f	$+0.70$
Annual rainfall, P	-0.48
Runoff coefficient, C	-0.28
TP of raw sewage, $\mathrm{TP}_{\mathrm{raw}}$	$+0.24$
TP of tertiary effluent, $\mathrm{TP}_{\mathrm{ter}}$	$+0.14$

Leakage is the single most important input. Even with tertiary treatment and less than 3% leakage, lake TP stays below 0.1 mg/L in only 31% of runs.

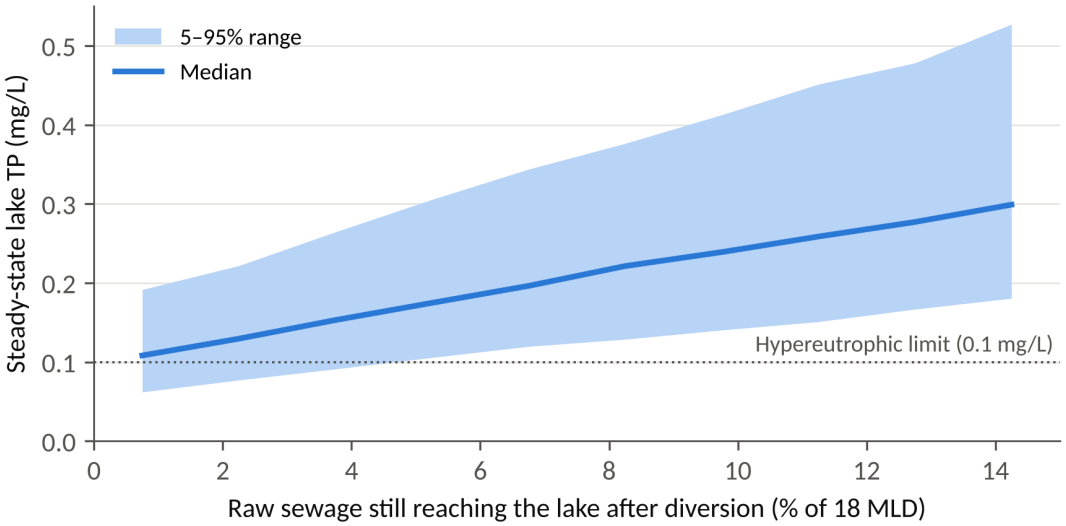


Figure 3: Lake TP versus residual sewage leakage, with tertiary effluent returned (scenario S2).

3.5 Lake-Problem parameters and lake type

After diversion the loss rate is $k = 3.9\,\text{yr}^{-1}$ (2.1 – 7.3). This gives:

$$b_{\text{annual}} = 1 - e^{-3.9} = \mathbf{0.98}, \qquad b_{\text{monthly}} = 1 - e^{-3.9/12} = \mathbf{0.28}\ \ (0.16\text{–}0.46)$$

For comparison, the classroom lake has $b = 0.42$, i.e. $k = -\ln(1 - 0.42) = 0.54\,\text{yr}^{-1}$, so Powai loses phosphorus about 7 times faster.

Table 8: Thresholds for lake type, and the sediment recycling strength r/m that Powai would need to reach each type ($k = 3.9\,\text{yr}^{-1}$).

q	$g_{\max}$	$f'_{\max}$	**Hysteretic if** $r/m >$	**Irreversible if** $r/m >$
2	0.500	0.650	6.0 yr^{-1}	7.8 yr^{-1}
3	0.529	0.840	4.6 yr^{-1}	7.4 yr^{-1}
4	0.570	1.065	3.7 yr^{-1}	6.8 yr^{-1}

- **Classroom lake:** $b = 0.42 < g_{\max} = 0.50$, so it is irreversible.
- **Powai:** it is reversible unless the sediments recycle phosphorus strongly (r/m above about 4–8 per year). How strong this recycling is decides whether Powai can recover, so it becomes a key uncertainty.

4. Difficulties encountered

- **Short phosphorus record:** only two short sampling campaigns (2016–17 and 2018–20) are public, which limits calibration. The IIT Bombay–BMC 12-month study [9] is not yet released.
- **Lake volume:** there is no bathymetry survey. One source quotes 840 million litres [10], while area × depth gives 5.8–10.7 Mm^3, so volume is kept as an uncertain input.

- **Unmeasured inputs:** sewage phosphorus and leakage after diversion have not been measured, so literature ranges were used.
- **Time step:** water stays in the lake for only about 0.5 yr, which is shorter than the annual model step ($b_{\text{annual}} \approx 0.98$). An annual model therefore forgets almost everything each year.

5. Changes to the proposed methodology

- Switch from an annual to a **monthly time step** ($b_{\text{monthly}} \approx 0.28$), to capture monsoon flushing and the dry-season build-up.
- Replace the abstract decision a_t with **explicit levers**: effluent TP standard, leakage control (sewer interception) and hyacinth harvesting.
- Add **leakage**, the **rainfall trend** and the **recycling parameters** (r/m, q) to the deeply uncertain factors.

6. Next steps

1. Build the monthly Carpenter–Vollenweider model, and narrow the ranges of r/m and q using the seasonal TP pattern (2018–20).
2. Run NSGA-II (pymoo) over the levers with three objectives: NPV utility (treatment cost versus damage), mean TP, and reliability of keeping TP < 0.1 mg/L.
3. Re-test the Pareto policies over about 1,000 Latin Hypercube futures, and use PRIM to find the conditions under which they fail.
4. Turn the robust policies into recommendations for the STP effluent standard and leakage control.

References

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